

Program-9/

```
|— namespaces.yaml
|— pods.yaml
```

Step 1: namespaces.yaml

```
apiVersion: v1
kind: Namespace
metadata:
  name: dev
```

```
apiVersion: v1
kind: Namespace
metadata:
  name: prod
```

Step 2: pods.yaml

```
apiVersion: v1
kind: Pod
metadata:
  name: frontend-pod
  namespace: dev
  labels:
    app: frontend
    env: dev
```

```
spec:
  containers:
  - name: frontend
    image: nginx
    ports:
    - containerPort: 80
```

```
apiVersion: v1
kind: Pod
metadata:
  name: backend-pod
  namespace: dev
  labels:
    app: backend
    env: dev
```

```
spec:
  containers:
  - name: backend
    image: nginx
    ports:
    - containerPort: 80
```

```
---
apiVersion: v1
kind: Pod
metadata:
  name: frontend-pod
  namespace: prod
  labels:
    app: frontend
    env: prod
spec:
  containers:
  - name: frontend
    image: nginx
    ports:
    - containerPort: 80
```

```
---
apiVersion: v1
kind: Pod
metadata:
  name: backend-pod
  namespace: prod
  labels:
    app: backend
    env: prod
spec:
  containers:
  - name: backend
    image: nginx
    ports:
    - containerPort: 80
```

Step 3: Apply YAML files

```
kubectl apply -f namespaces.yaml
kubectl apply -f pods.yaml
```

step 4: Verify Pods by Namespace

```
kubectl get pods -n dev
kubectl get pods -n prod
```

Step 5: Filter Pods by Label

List frontend pods in dev

```
kubectl get pods -n dev -l app=frontend
```

List backend pods in prod

```
kubectl get pods -n prod -l app=backend
```

output

```
1RV24MC096_SAURABHR@sentient:~/Docker_MCA -/demo9
1RV24MC096_SAURABHR@sentient:~/Docker_MCA$ kubectl apply -f namespaces.yaml
error: error validating "namespaces.yaml": error validating data: failed to download openapi: Get "https://192.168.49.2:8443/openapi/v2?timeout=32s": dial tcp 192.168.49.2:8443: connect: no route to host: if you choose to ignore these errors, turn validation off with --validate=false
1RV24MC096_SAURABHR@sentient:~/Docker_MCA$ minikube start
🐳 minikube v1.37.0 on Ubuntu 24.04
🔗 Using the docker driver based on existing profile
👉 Starting "minikube" primary control-plane node in "minikube" cluster
📶 Pulling base image v0.0.48 ...
🔄 Restarting existing docker container for "minikube" ...
❗ Failing to connect to https://registry.k8s.io/ from inside the minikube container
💡 To pull new external images, you may need to configure a proxy: https://minikube.sigs.k8s.io/docs/reference/networking/proxy/
🔄 Preparing Kubernetes v1.34.0 on Docker 28.4.0 ...
🔍 Verifying Kubernetes components...
   ▪ Using image gcr.io/k8s-minikube/storage-provisioner:v5
🌟 Enabled addons: default-storageclass, storage-provisioner
🏠 Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
1RV24MC096_SAURABHR@sentient:~/Docker_MCA$ kubectl apply -f namespaces.yaml
namespace/dev created
namespace/prod created
1RV24MC096_SAURABHR@sentient:~/Docker_MCA$ kubectl apply -f pods.yaml
pod/frontend-pod created
pod/backend-pod created
pod/frontend-pod created
pod/backend-pod created
1RV24MC096_SAURABHR@sentient:~/Docker_MCA$ kubectl get pods -n dev
NAME          READY   STATUS    RESTARTS   AGE
backend-pod   1/1     Running   0           17s
frontend-pod  1/1     Running   0           17s
1RV24MC096_SAURABHR@sentient:~/Docker_MCA$ kubectl get pods -n prod
NAME          READY   STATUS    RESTARTS   AGE
backend-pod   1/1     Running   0           88s
frontend-pod  1/1     Running   0           88s
1RV24MC096_SAURABHR@sentient:~/Docker_MCA$ kubectl get pods
NAME          READY   STATUS    RESTARTS   AGE
kubla         1/1     Running   3 (3m57s ago)    8d
nginx-deployment-6f9664446b-48ff8  1/1     Running   4 (6d17s ago)    23d
nginx-deployment-6f9664446b-nfcxw  1/1     Running   4 (6d17s ago)    28d
nodejs-pod    1/1     Running   1 (3m57s ago)    6d18h
1RV24MC096_SAURABHR@sentient:~/Docker_MCA$ kubectl get pods -n dev -l app=frontend
NAME          READY   STATUS    RESTARTS   AGE
frontend-pod  1/1     Running   0           187s
1RV24MC096_SAURABHR@sentient:~/Docker_MCA$ kubectl get pods -n prod -l app=backend
NAME          READY   STATUS    RESTARTS   AGE
backend-pod   1/1     Running   0           2m1s
```